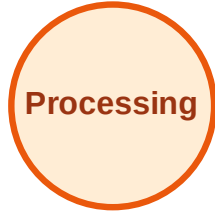


BFS Traversal

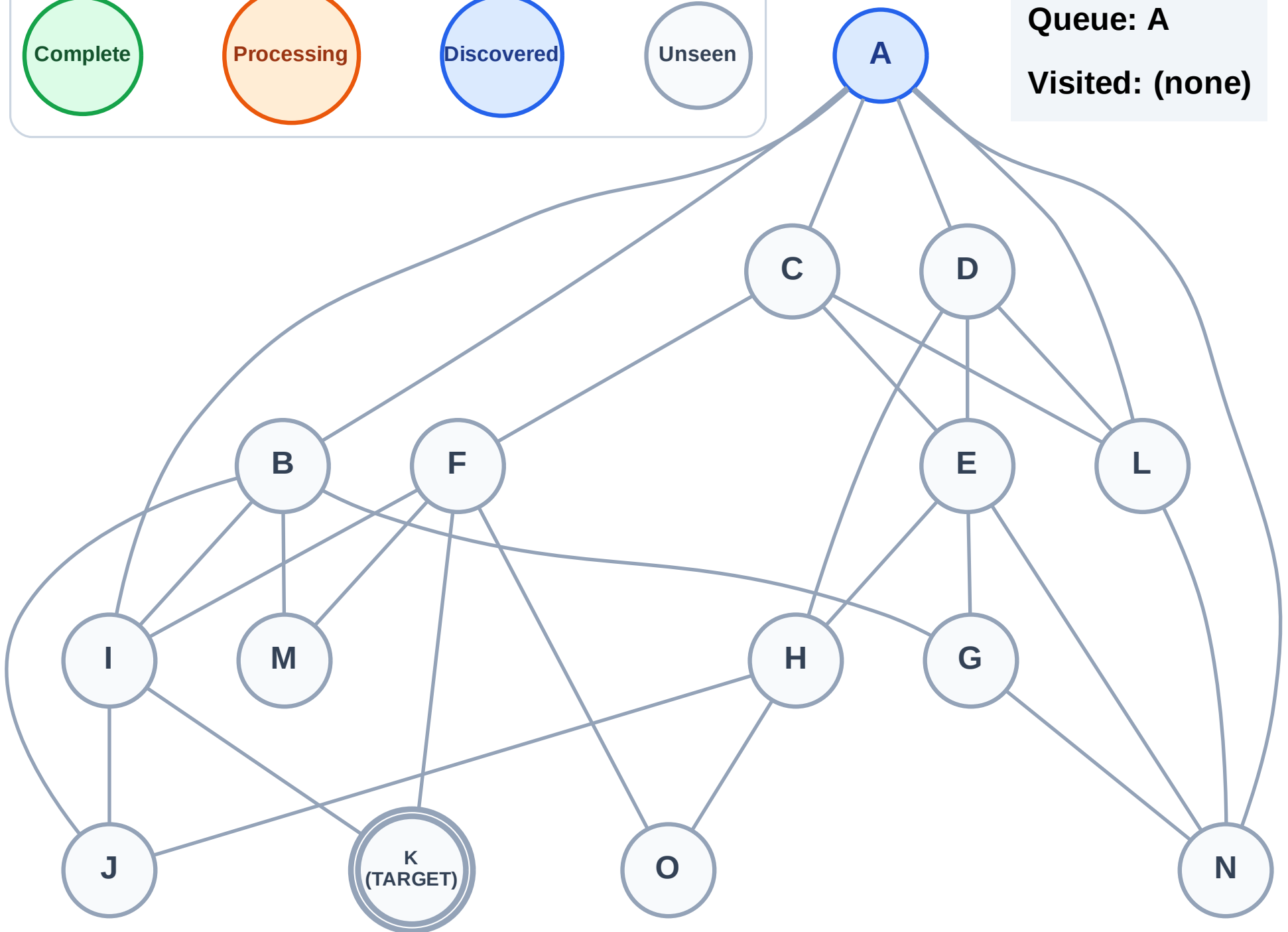
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

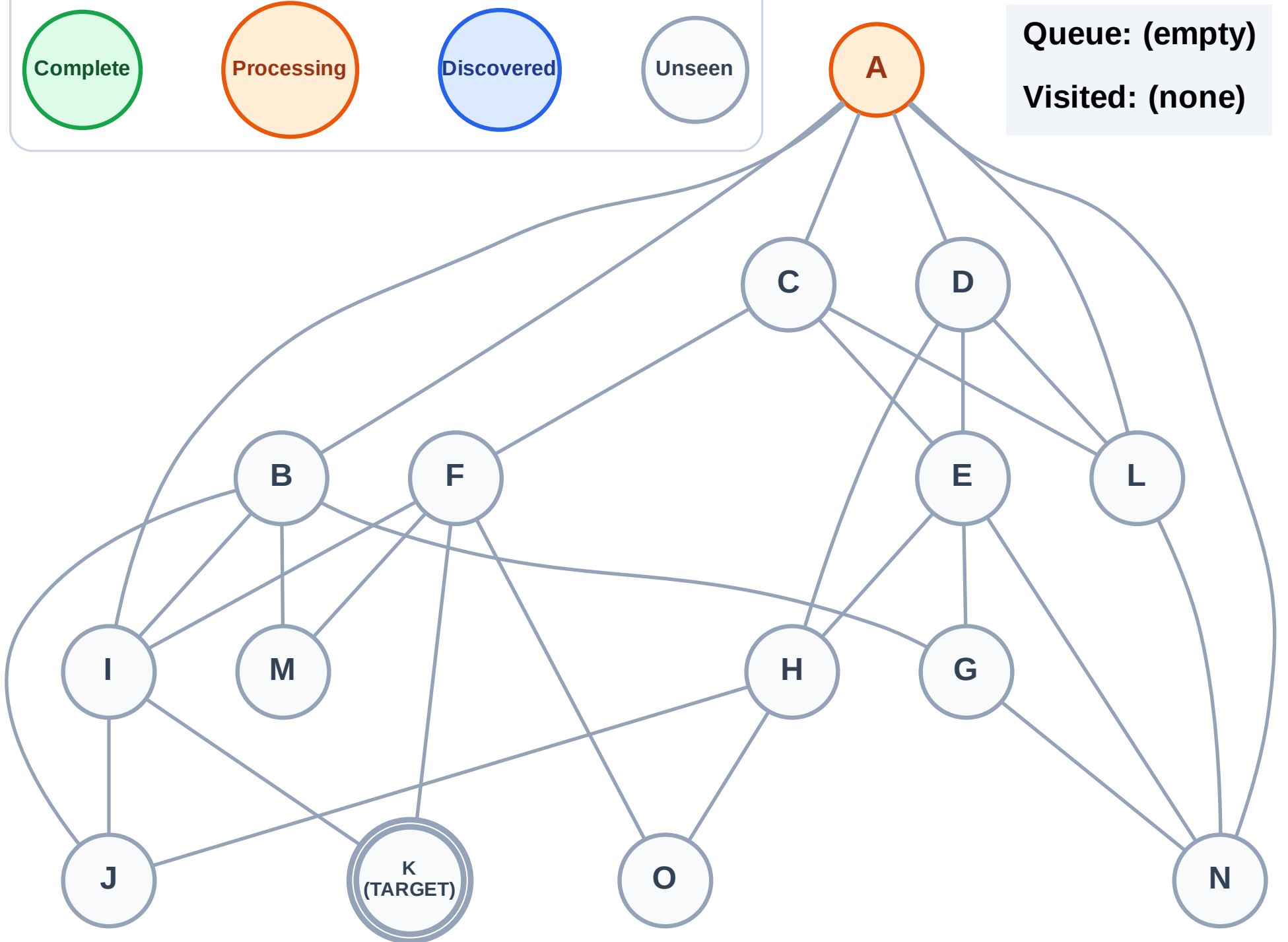
Step 2: Process node A

Legend



Queue: (empty)

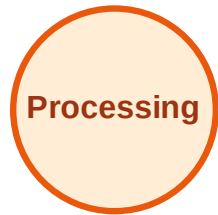
Visited: (none)



BFS Traversal

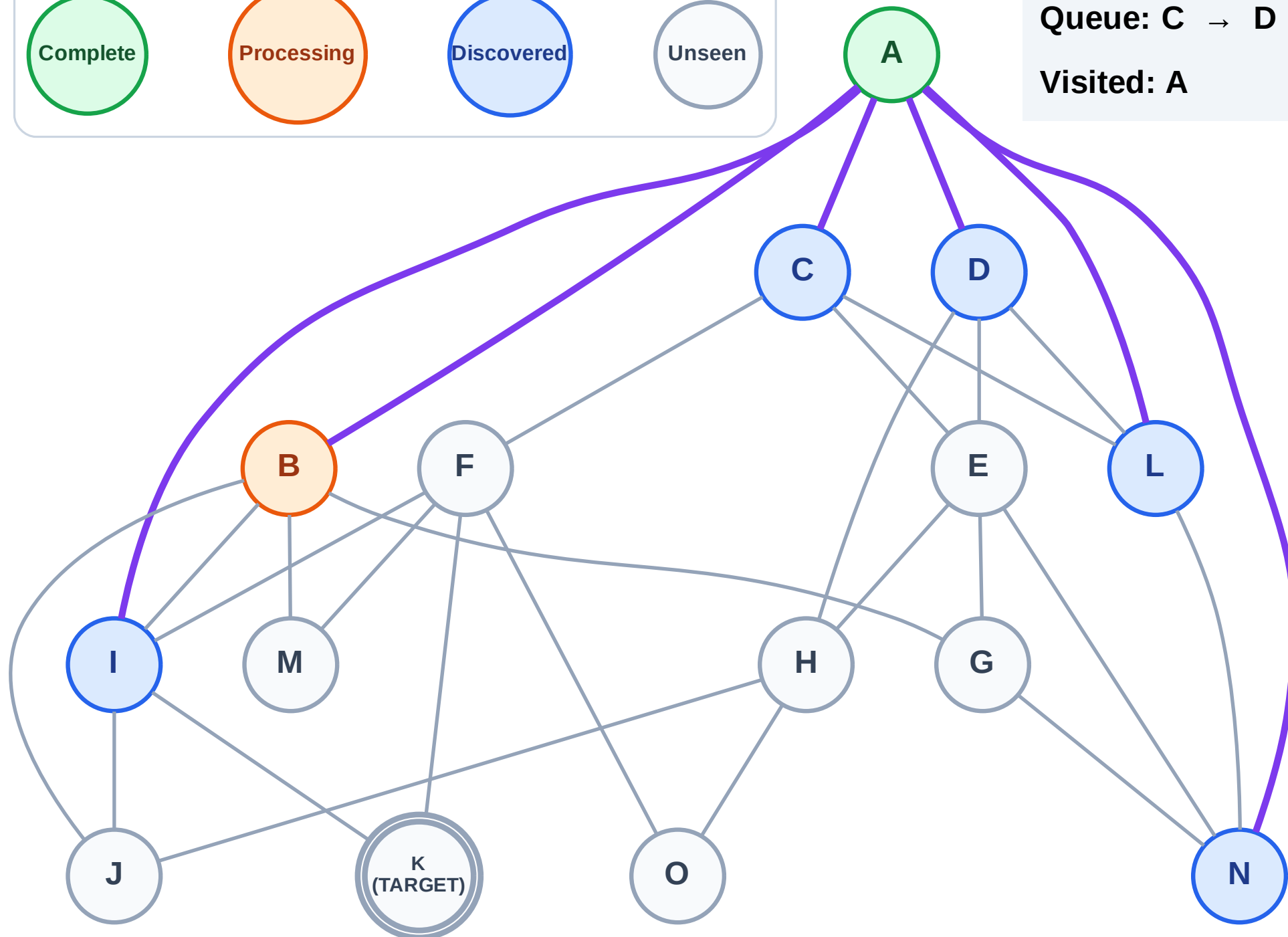
Step 4: Process node B

Legend



Queue: C → D → I → L → N

Visited: A



BFS Traversal

Step 5: Discover G, J, M from B

Legend

Complete

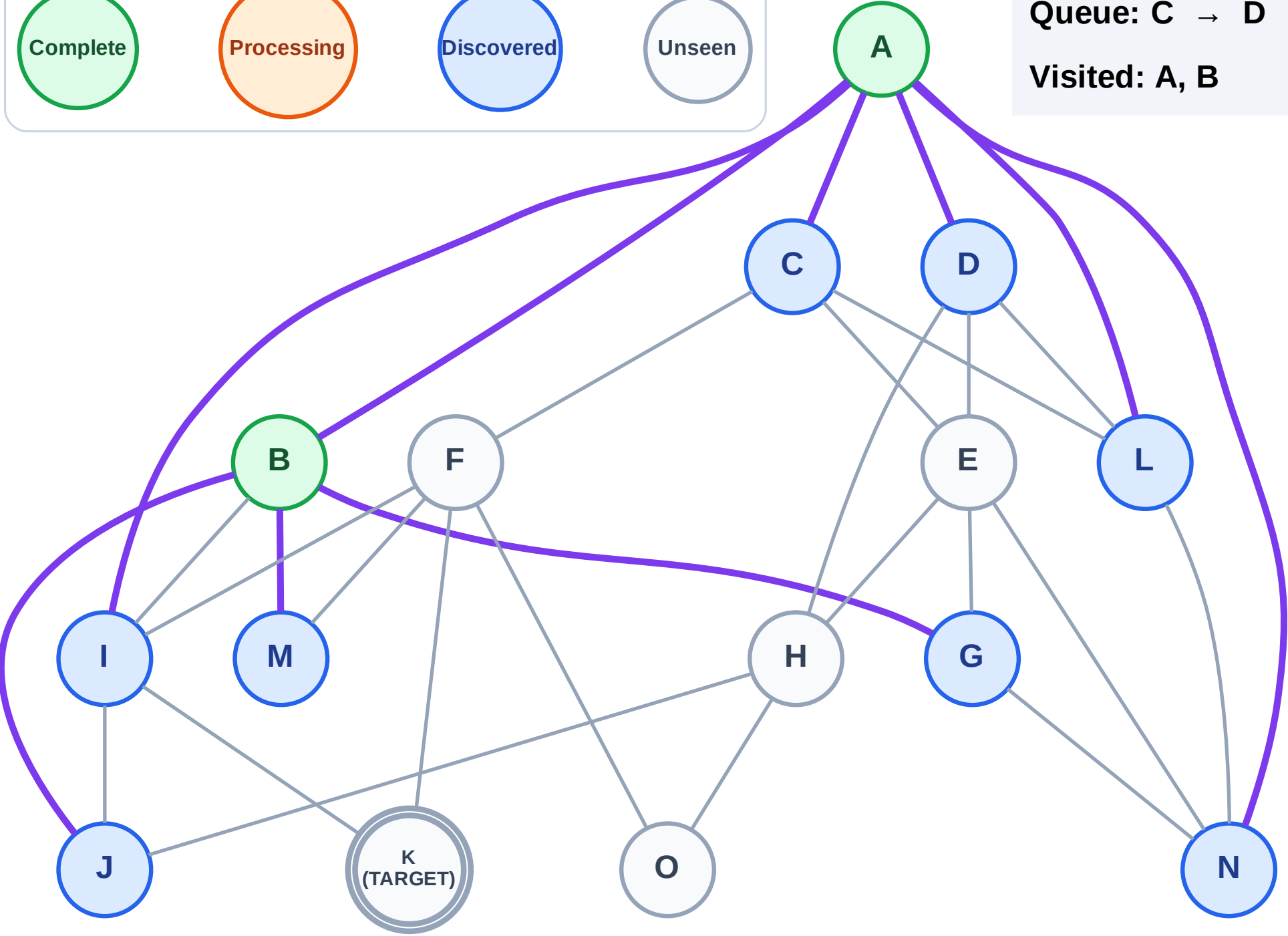
Processing

Discovered

Unseen

Queue: C → D → I → L → N → G → J → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

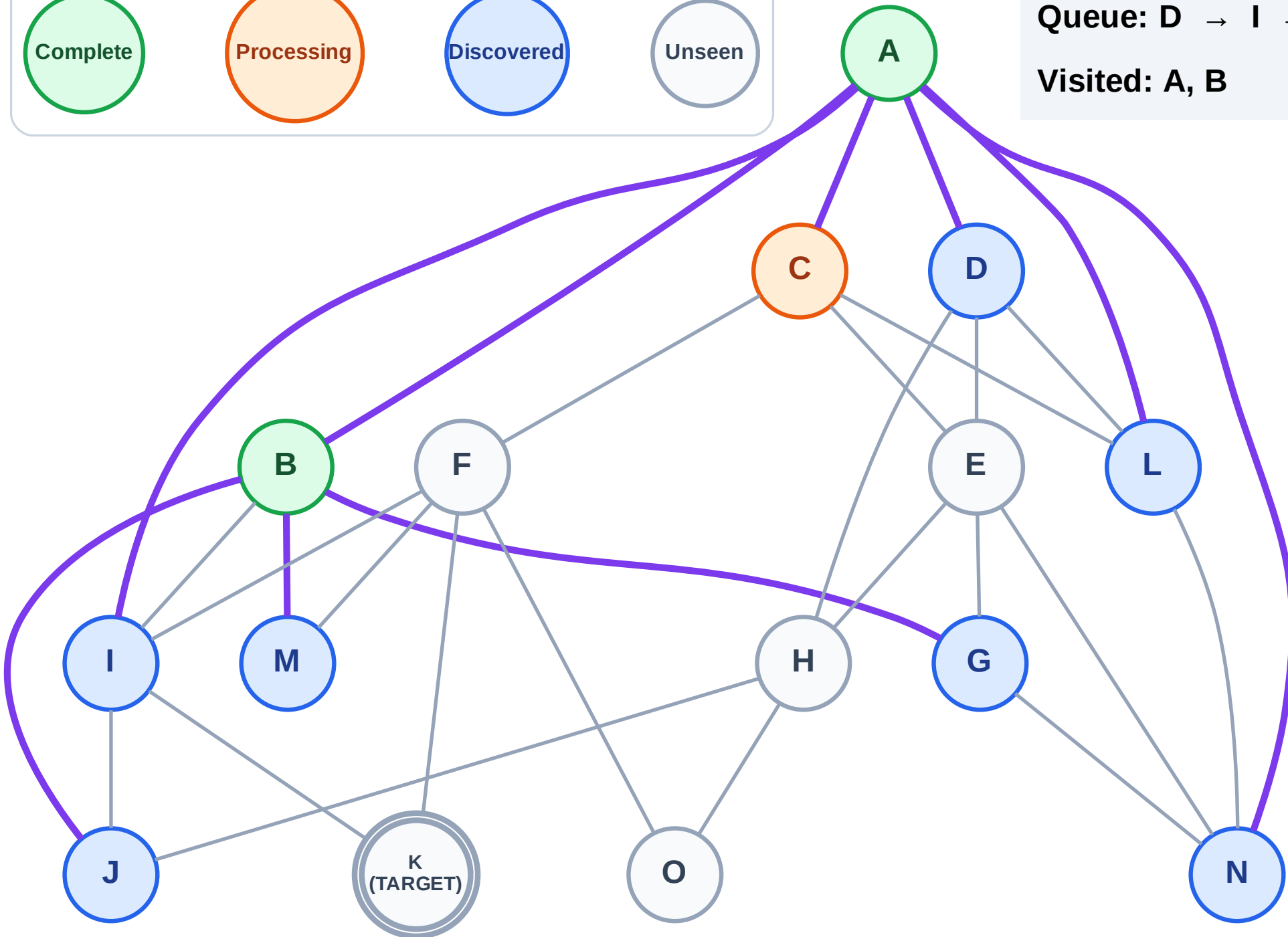
Processing

Discovered

Unseen

Queue: D → I → L → N → G → J → M

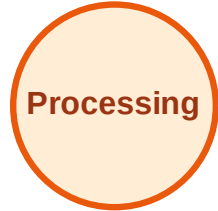
Visited: A, B



BFS Traversal

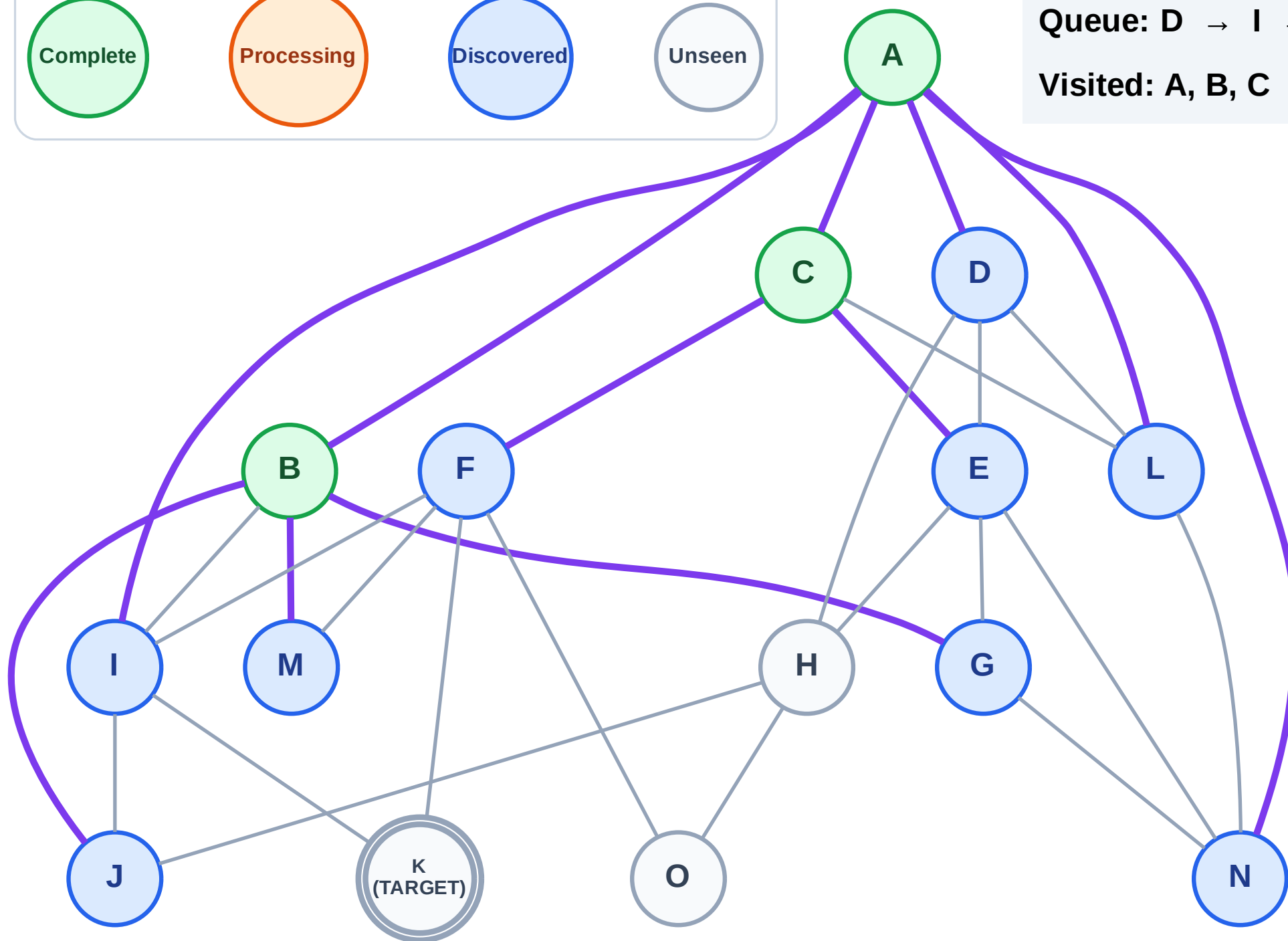
Step 7: Discover E, F from C

Legend



Queue: D → I → L → N → G → J → M → E → F

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

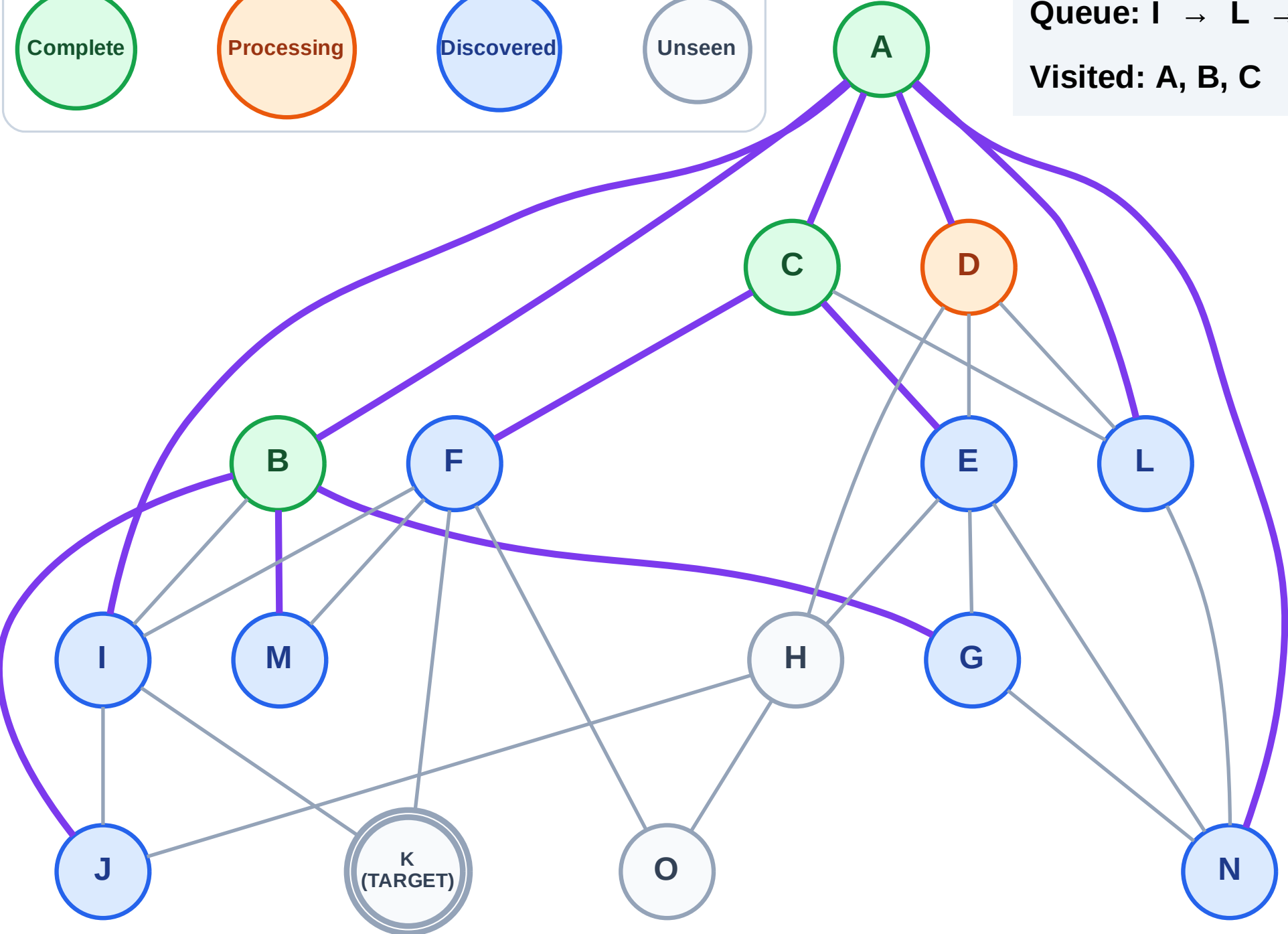
Processing

Discovered

Unseen

Queue: I → L → N → G → J → M → E → F

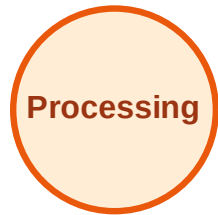
Visited: A, B, C



BFS Traversal

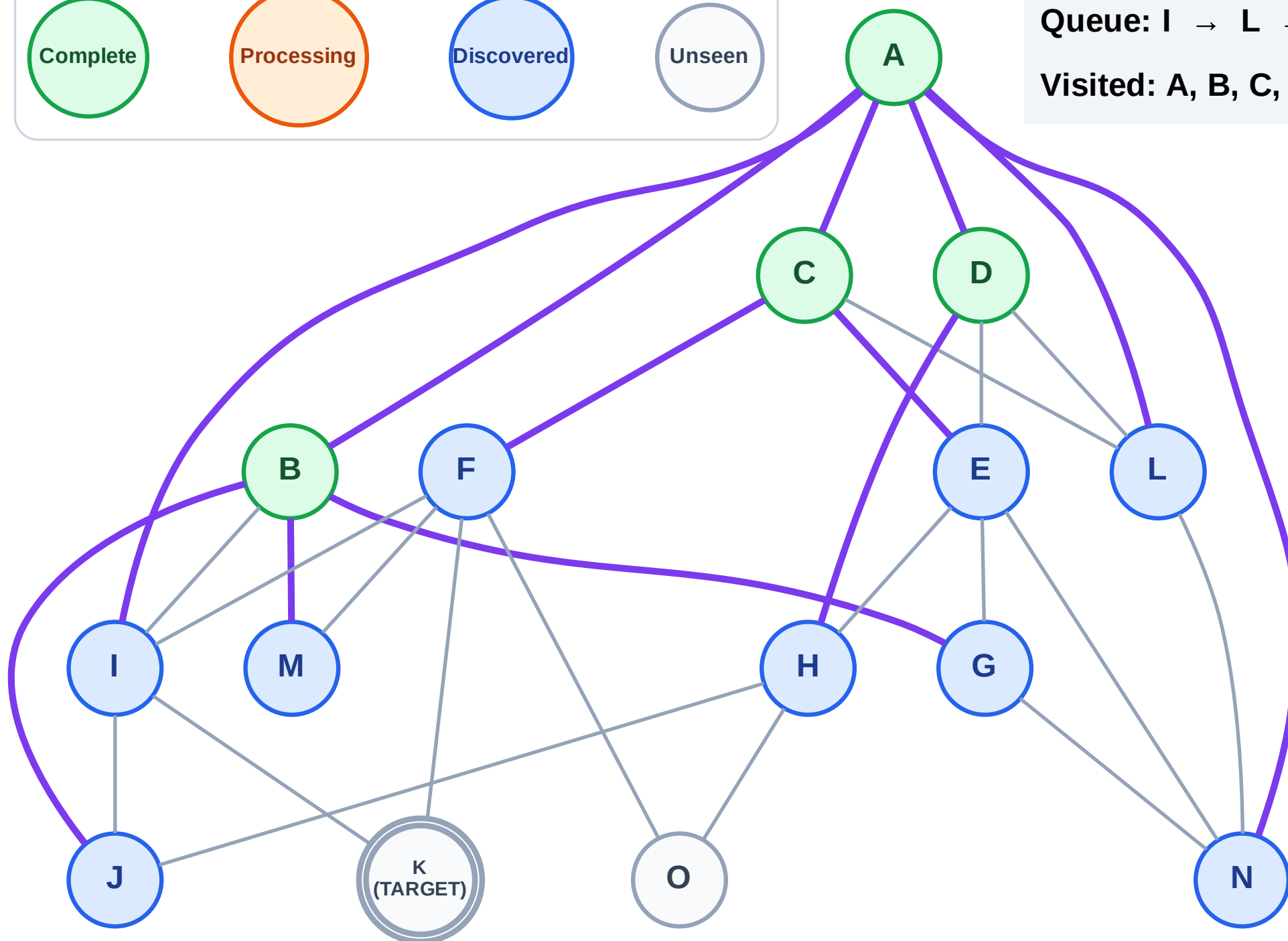
Step 9: Discover H from D

Legend



Queue: I → L → N → G → J → M → E → F → H

Visited: A, B, C, D



BFS Traversal

Step 10: Process node I

Legend

Complete

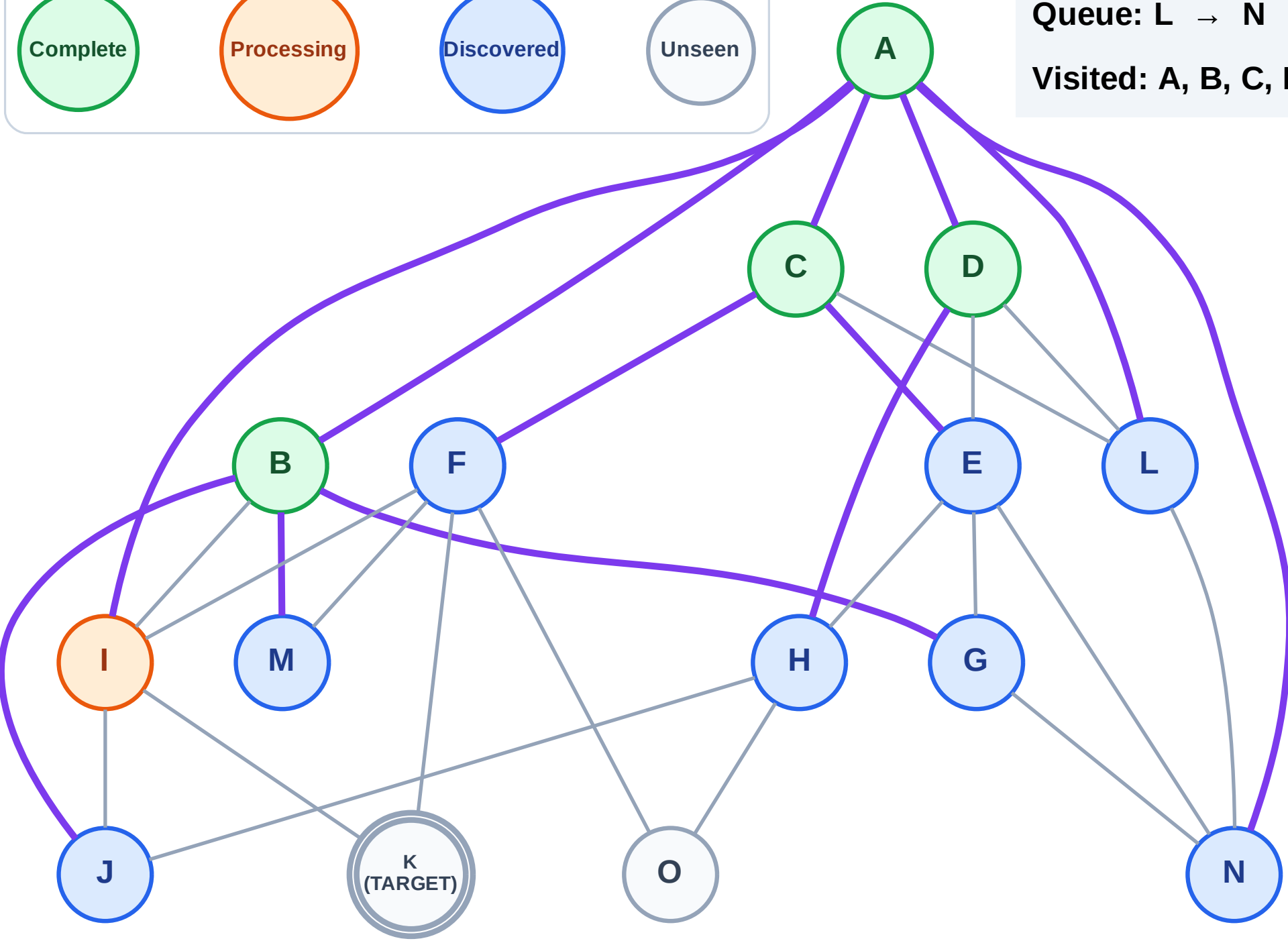
Processing

Discovered

Unseen

Queue: L → N → G → J → M → E → F → H

Visited: A, B, C, D



BFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → I → K

Legend

Complete

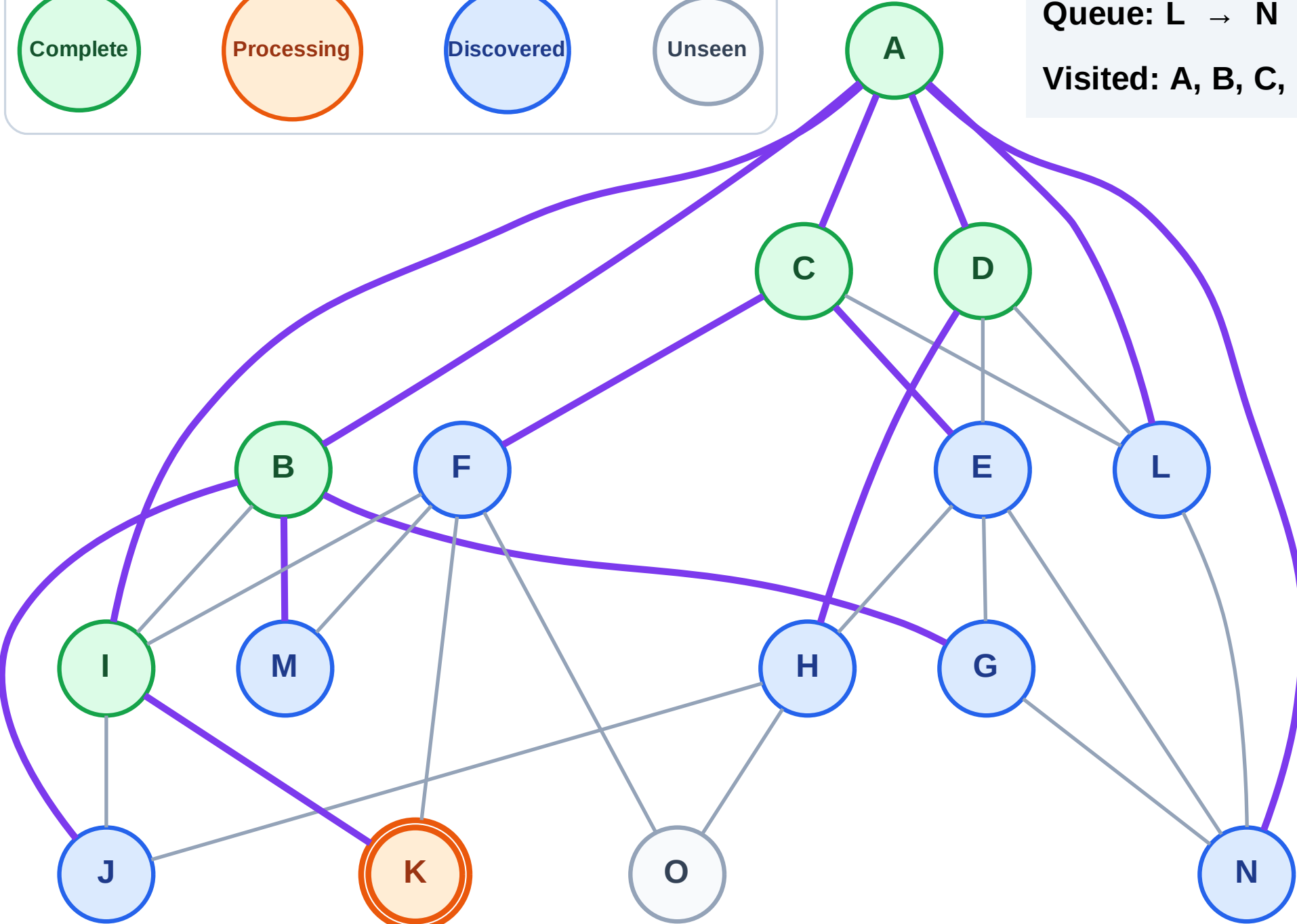
Processing

Discovered

Unseen

Queue: L → N → G → J → M → E → F → H

Visited: A, B, C, D, I, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

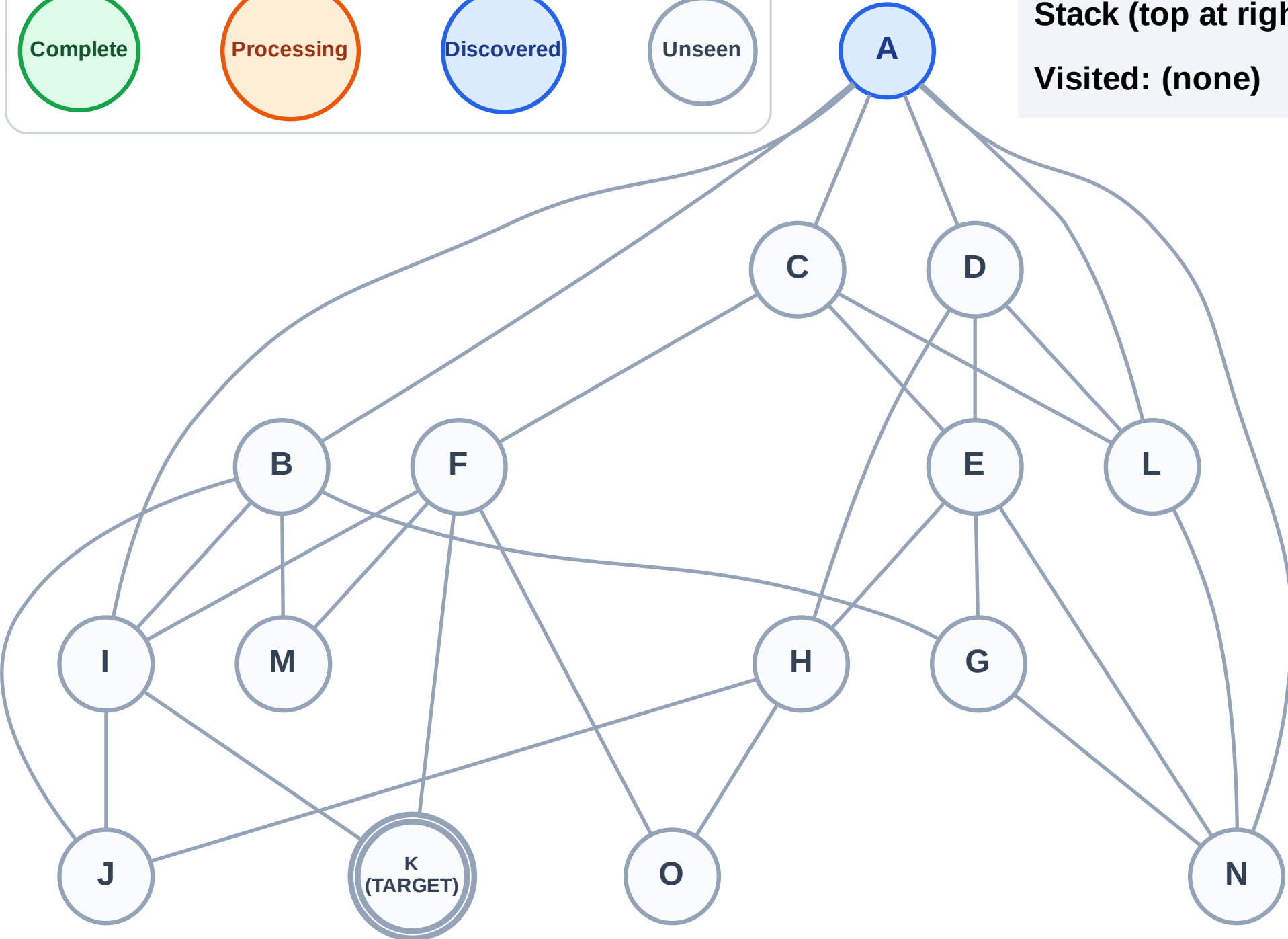
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

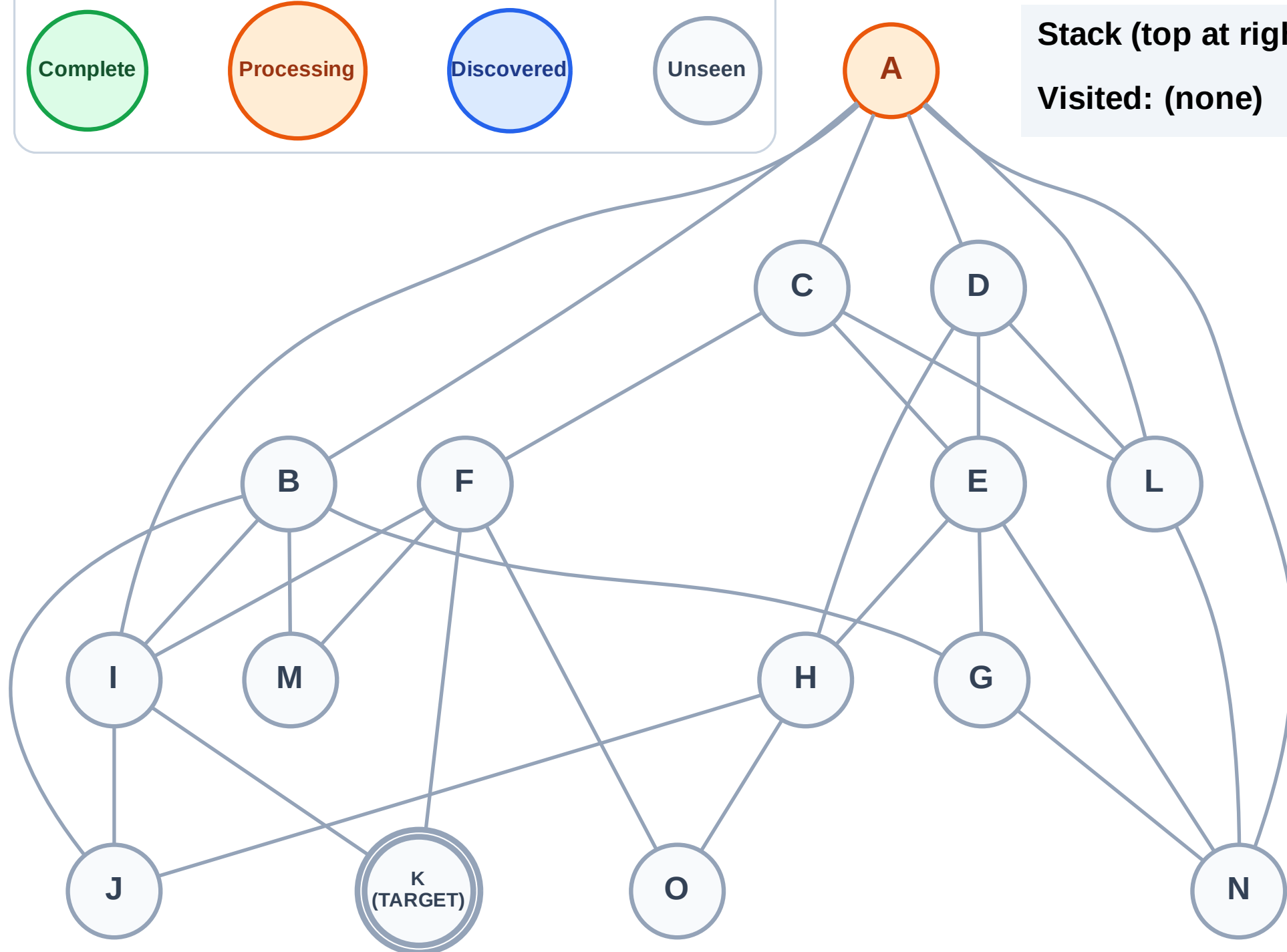
Processing

Discovered

Unseen

Stack (top at right): (empty)

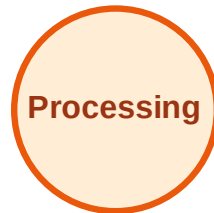
Visited: (none)



DFS Traversal

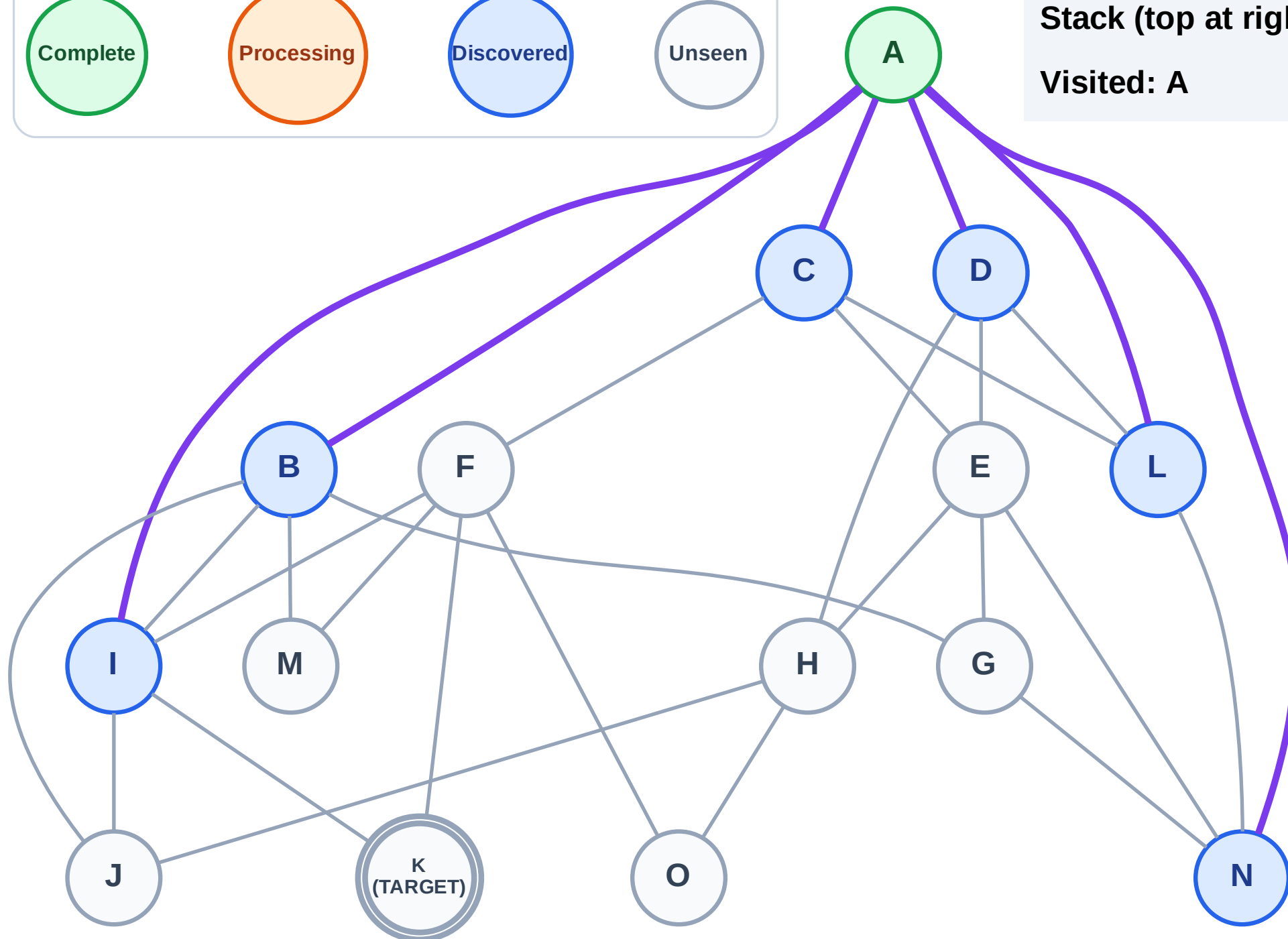
Step 3: Discover N, L, I, D, C, B from A

Legend



Stack (top at right): N | L | I | D | C | B

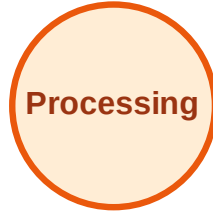
Visited: A



DFS Traversal

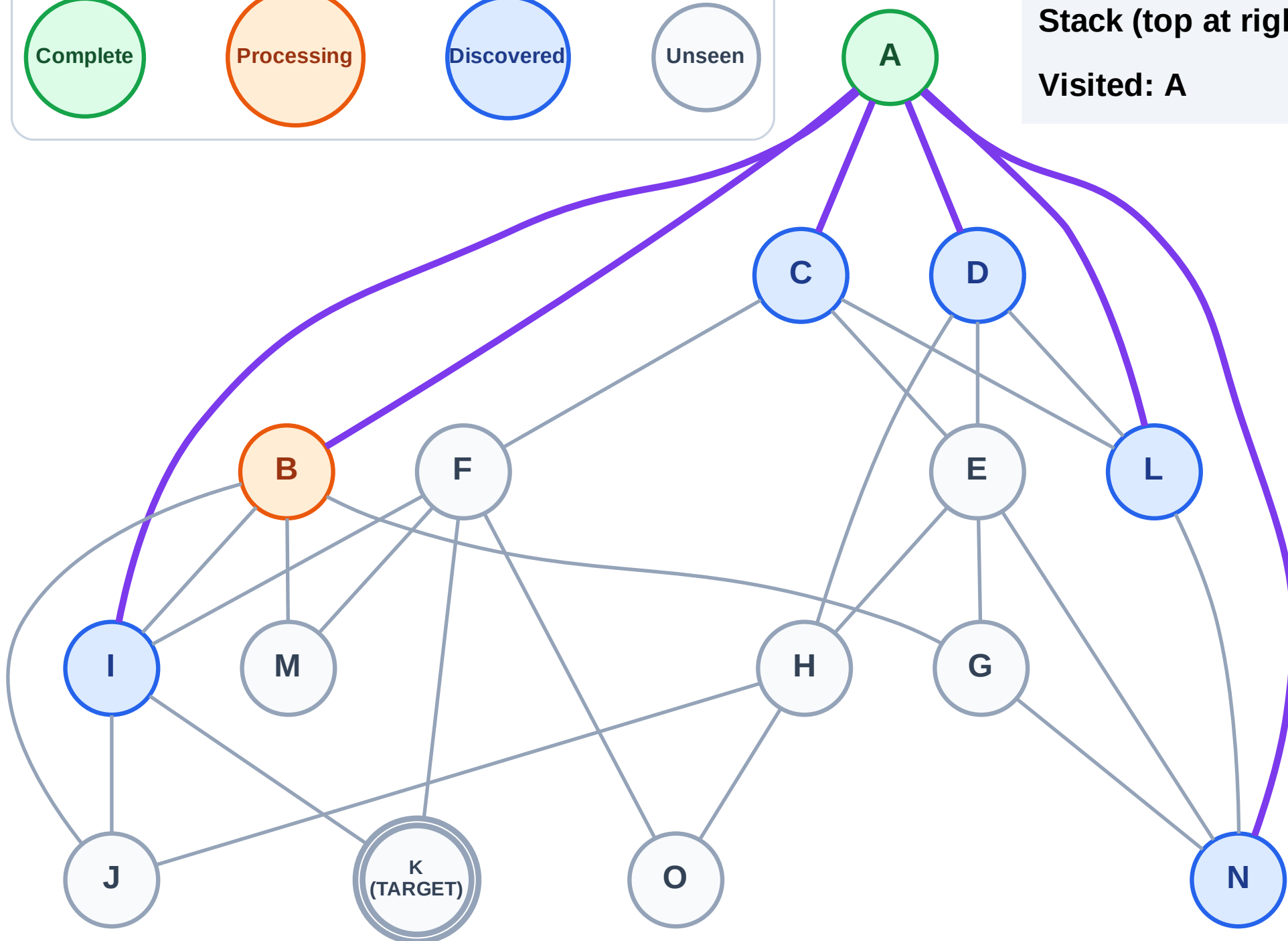
Step 4: Process node B

Legend



Stack (top at right): N | L | I | D | C

Visited: A



DFS Traversal

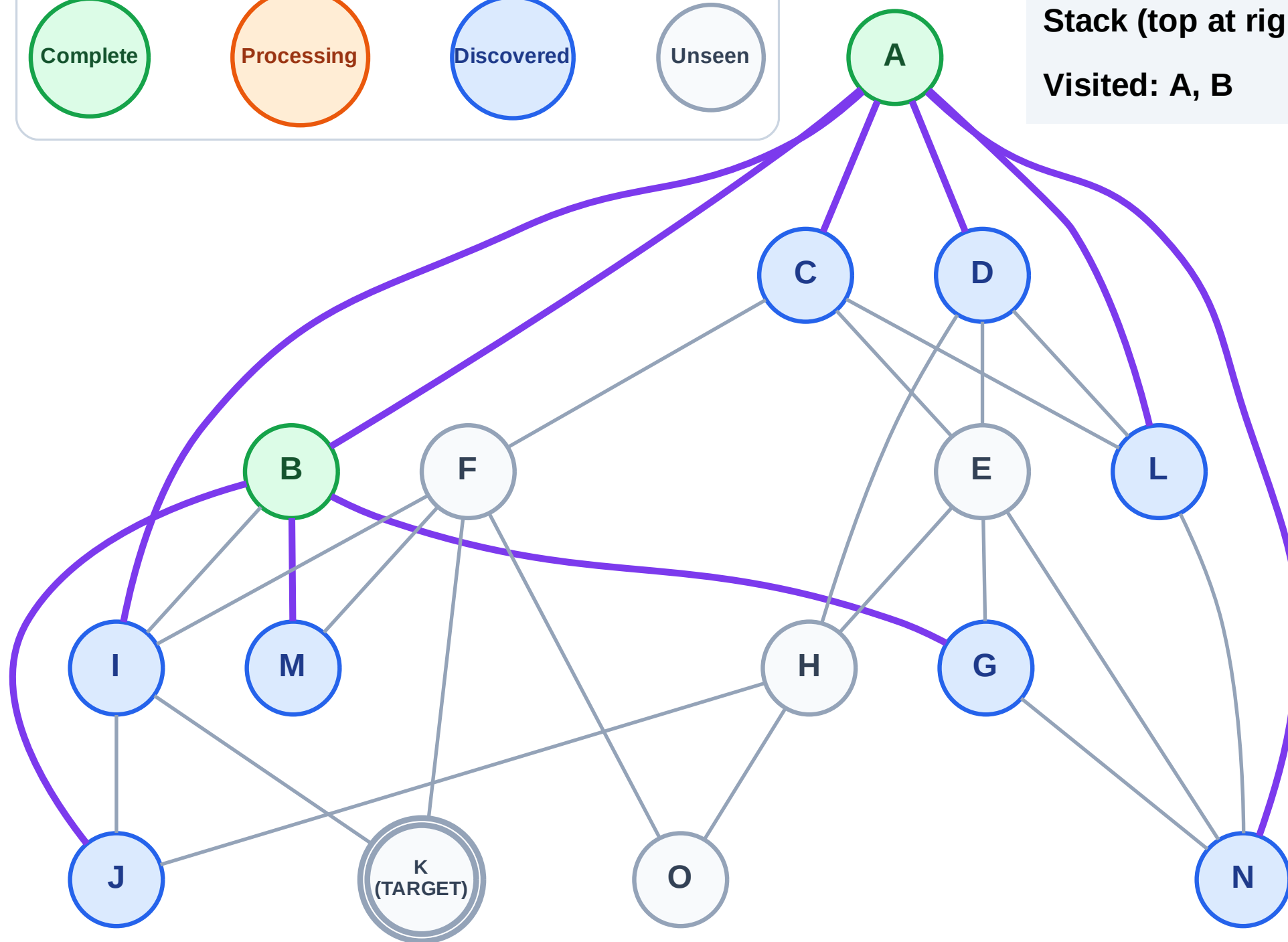
Step 5: Discover M, J, G from B

Legend



Stack (top at right): N | L | I | D | C | M | J | G

Visited: A, B



DFS Traversal

Step 6: Process node G

Legend

Complete

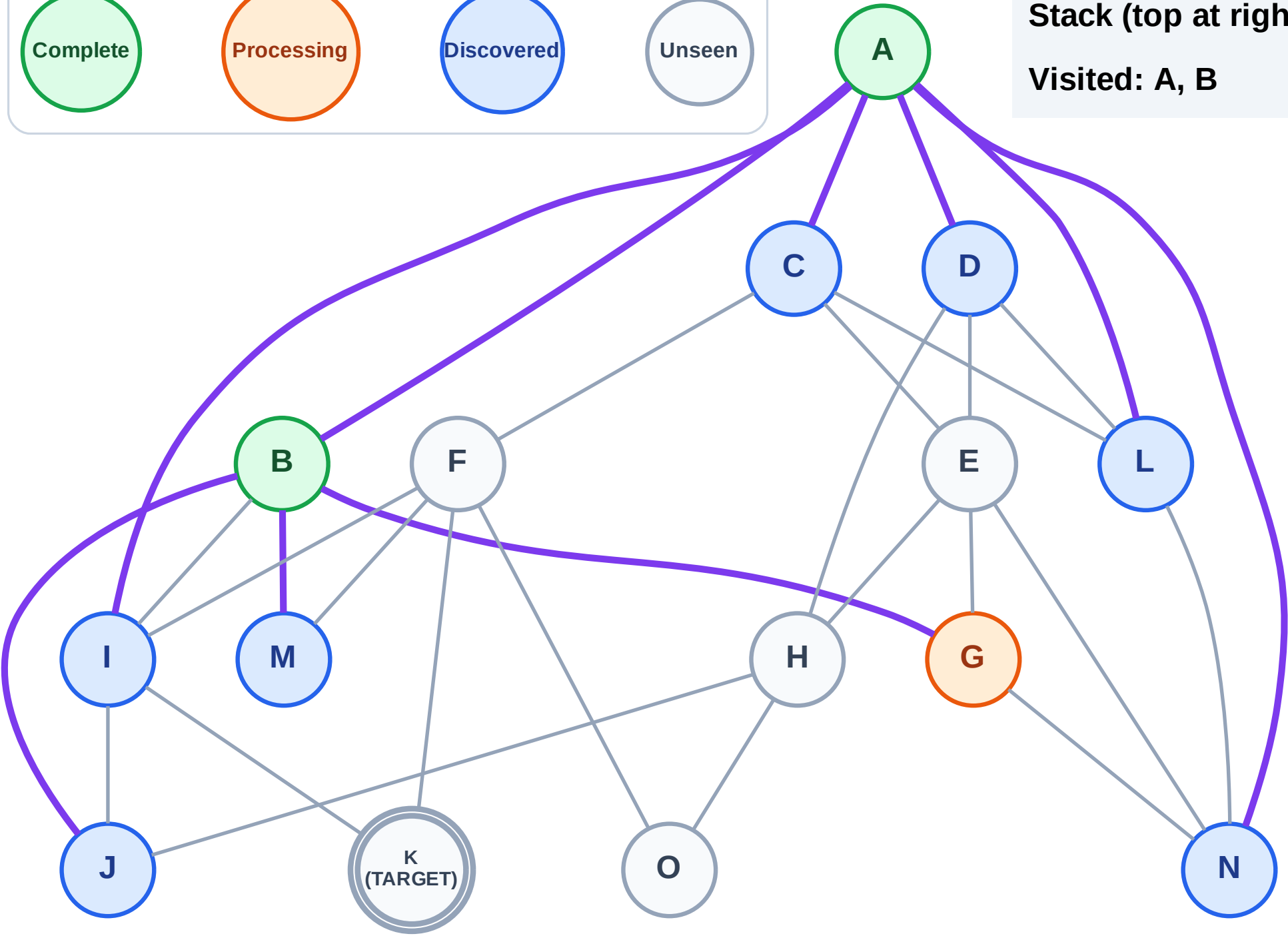
Processing

Discovered

Unseen

Stack (top at right): N | L | I | D | C | M | J

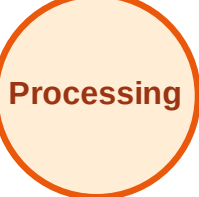
Visited: A, B



DFS Traversal

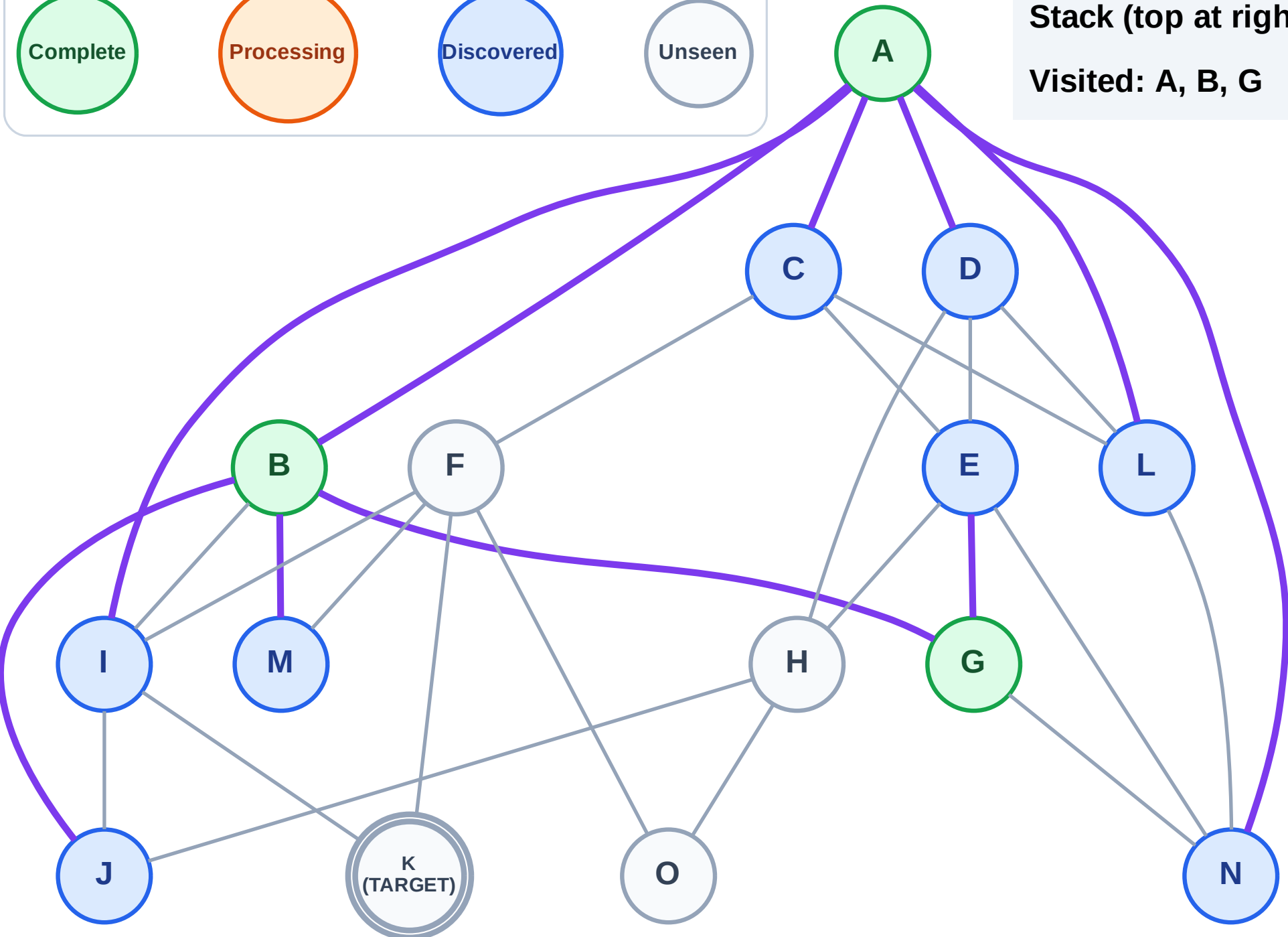
Step 7: Discover E from G

Legend



Stack (top at right): N | L | I | D | C | M | J | E

Visited: A, B, G



DFS Traversal

Step 8: Process node E

Legend

Complete

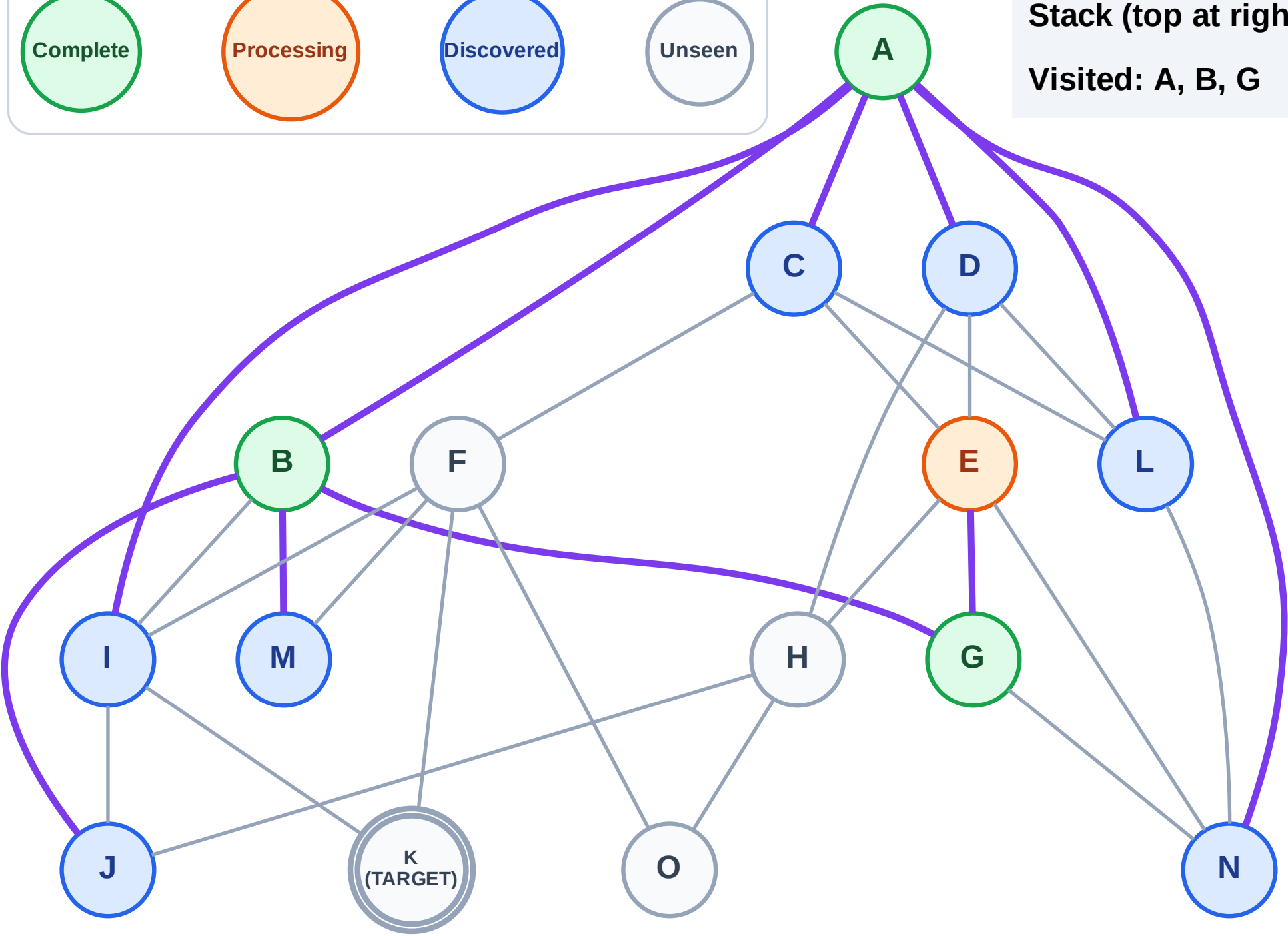
Processing

Discovered

Unseen

Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, G



DFS Traversal

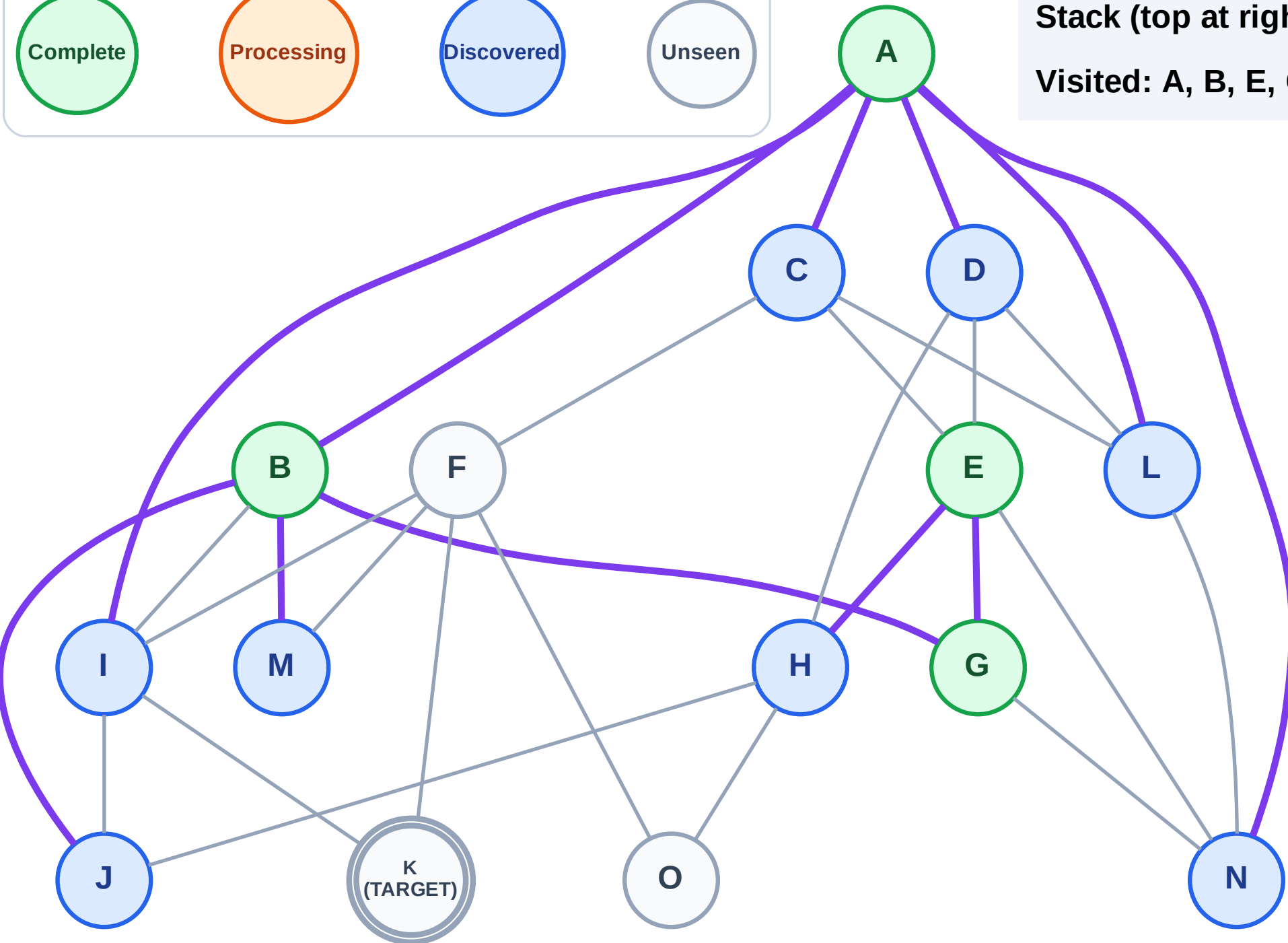
Step 9: Discover H from E

Legend



Stack (top at right): N | L | I | D | C | M | J | H

Visited: A, B, E, G



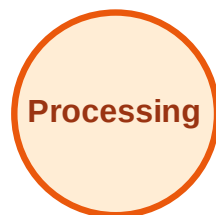
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



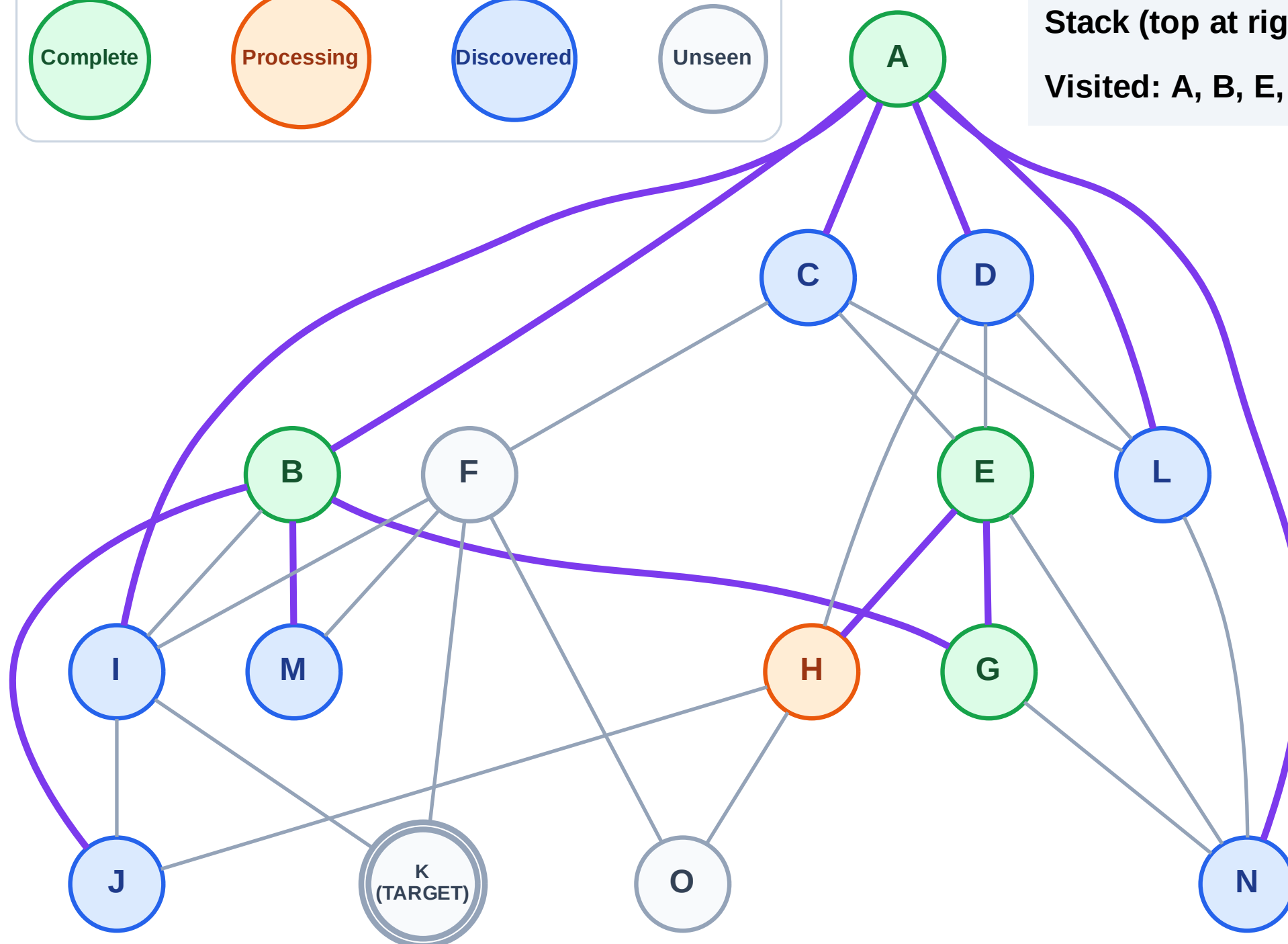
Discovered



Unseen

Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, E, G



DFS Traversal

Step 11: Discover O from H

Legend

Complete

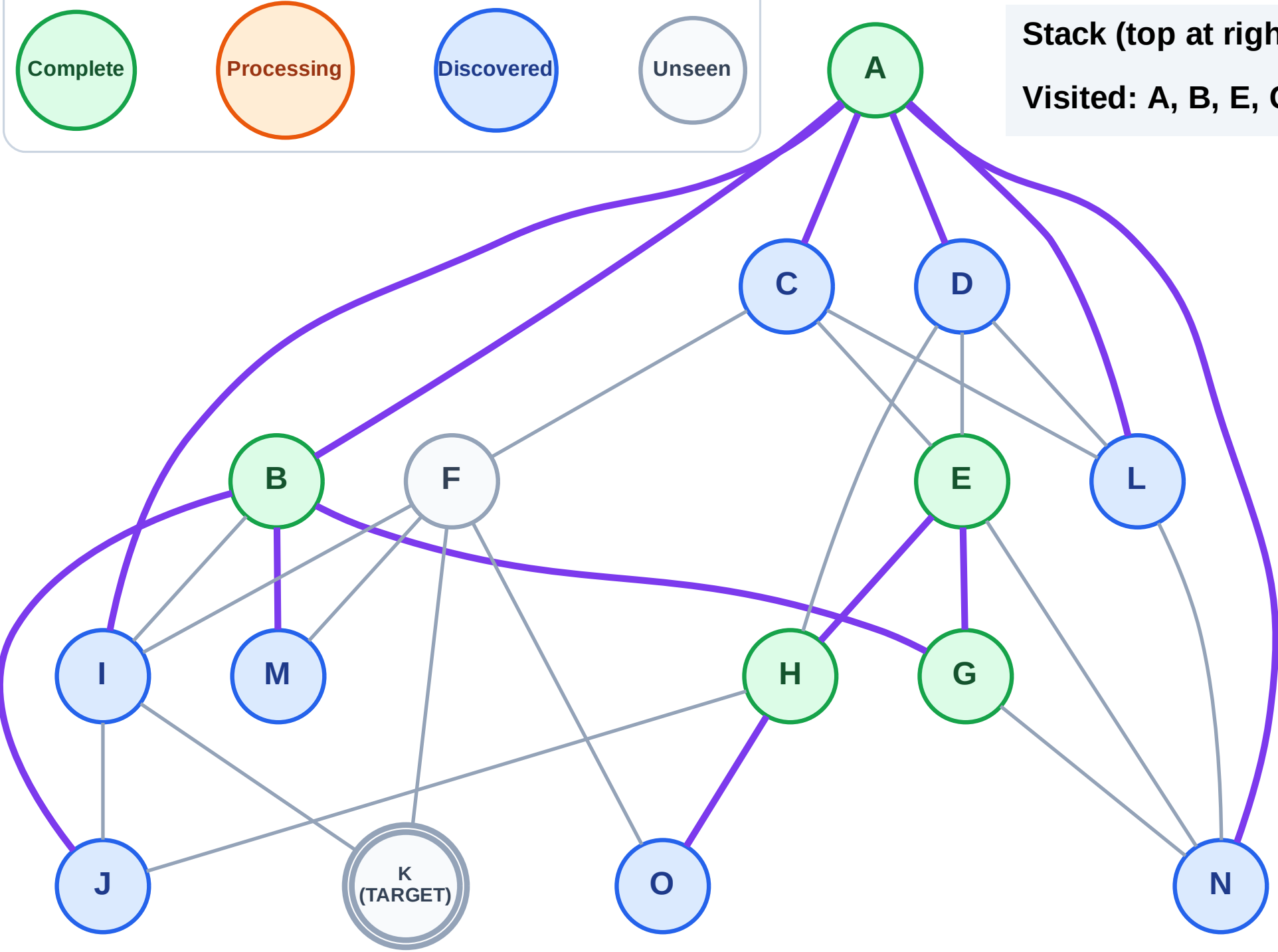
Processing

Discovered

Unseen

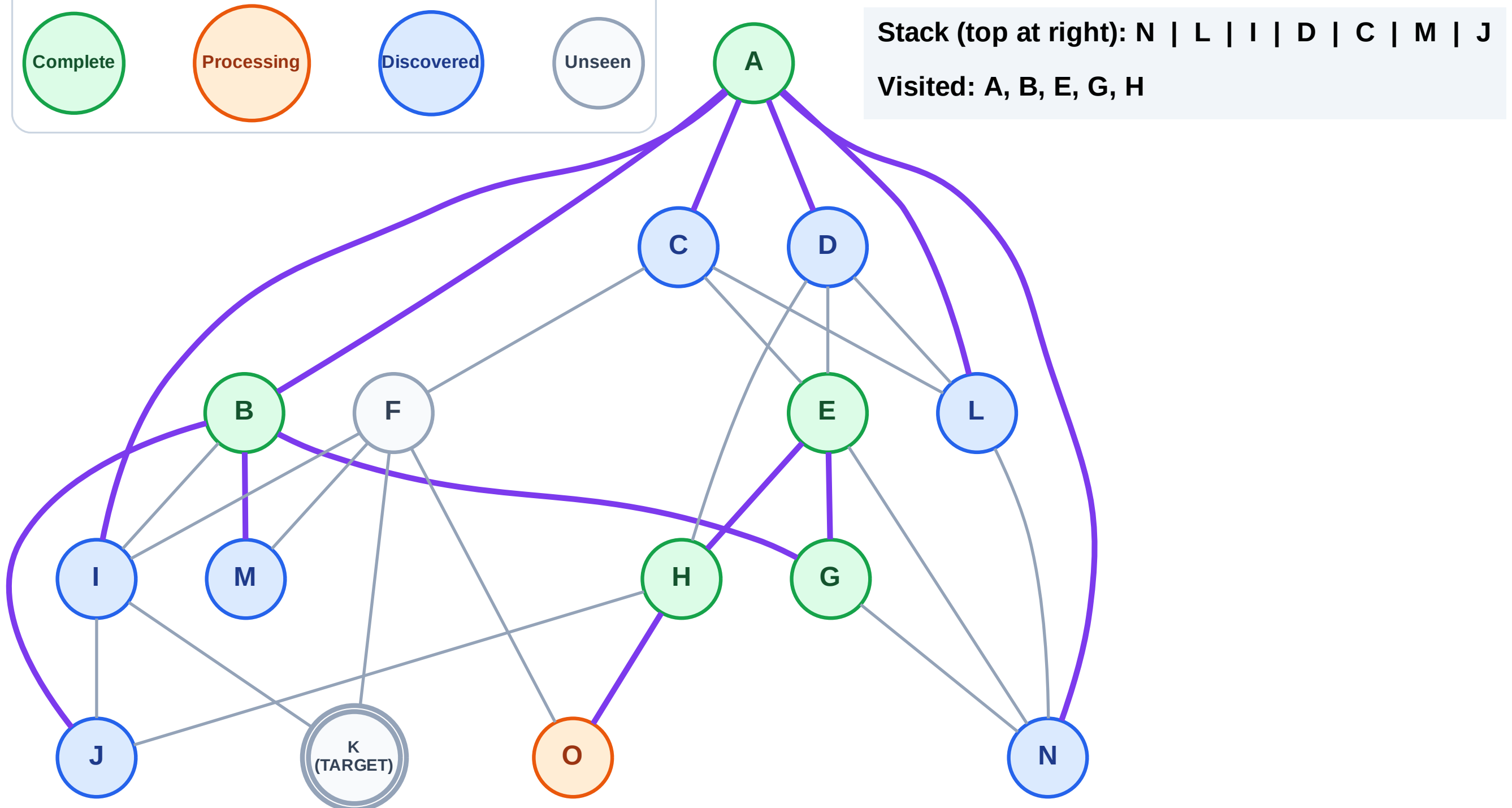
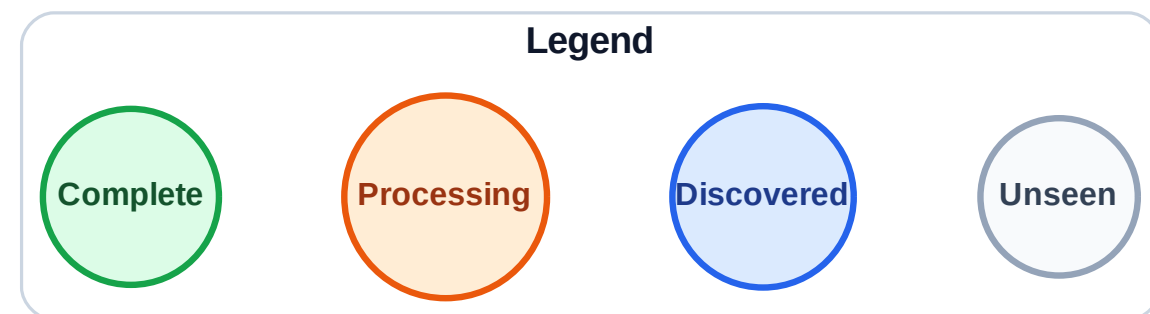
Stack (top at right): N | L | I | D | C | M | J | O

Visited: A, B, E, G, H



DFS Traversal

Step 12: Process node O



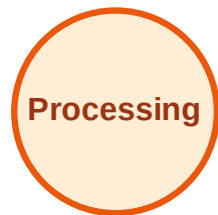
DFS Traversal

Step 13: Discover F from O

Legend



Complete



Processing



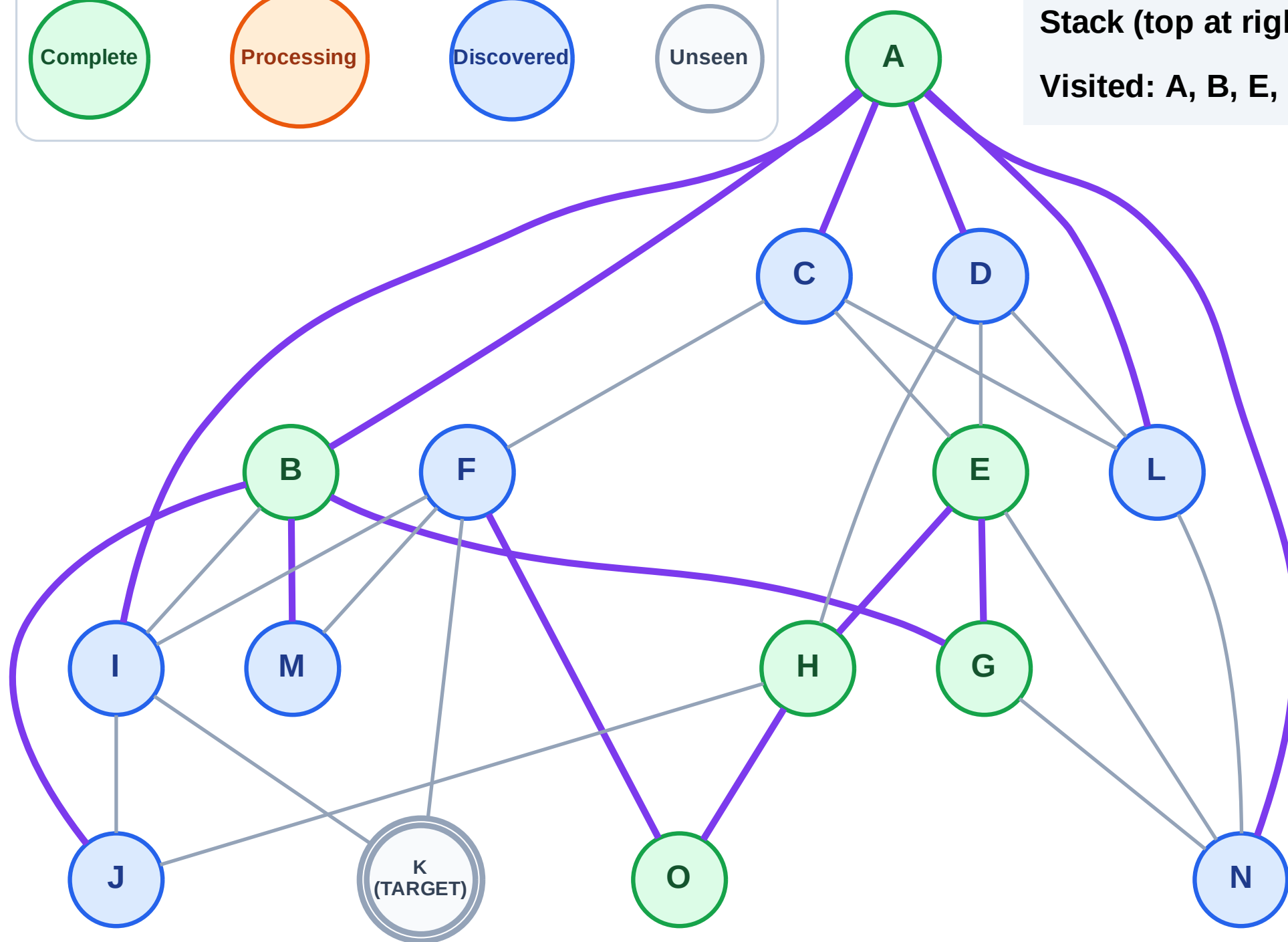
Discovered



Unseen

Stack (top at right): N | L | I | D | C | M | J | F

Visited: A, B, E, G, H, O



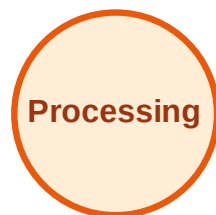
DFS Traversal

Step 14: Process node F

Legend



Complete



Processing



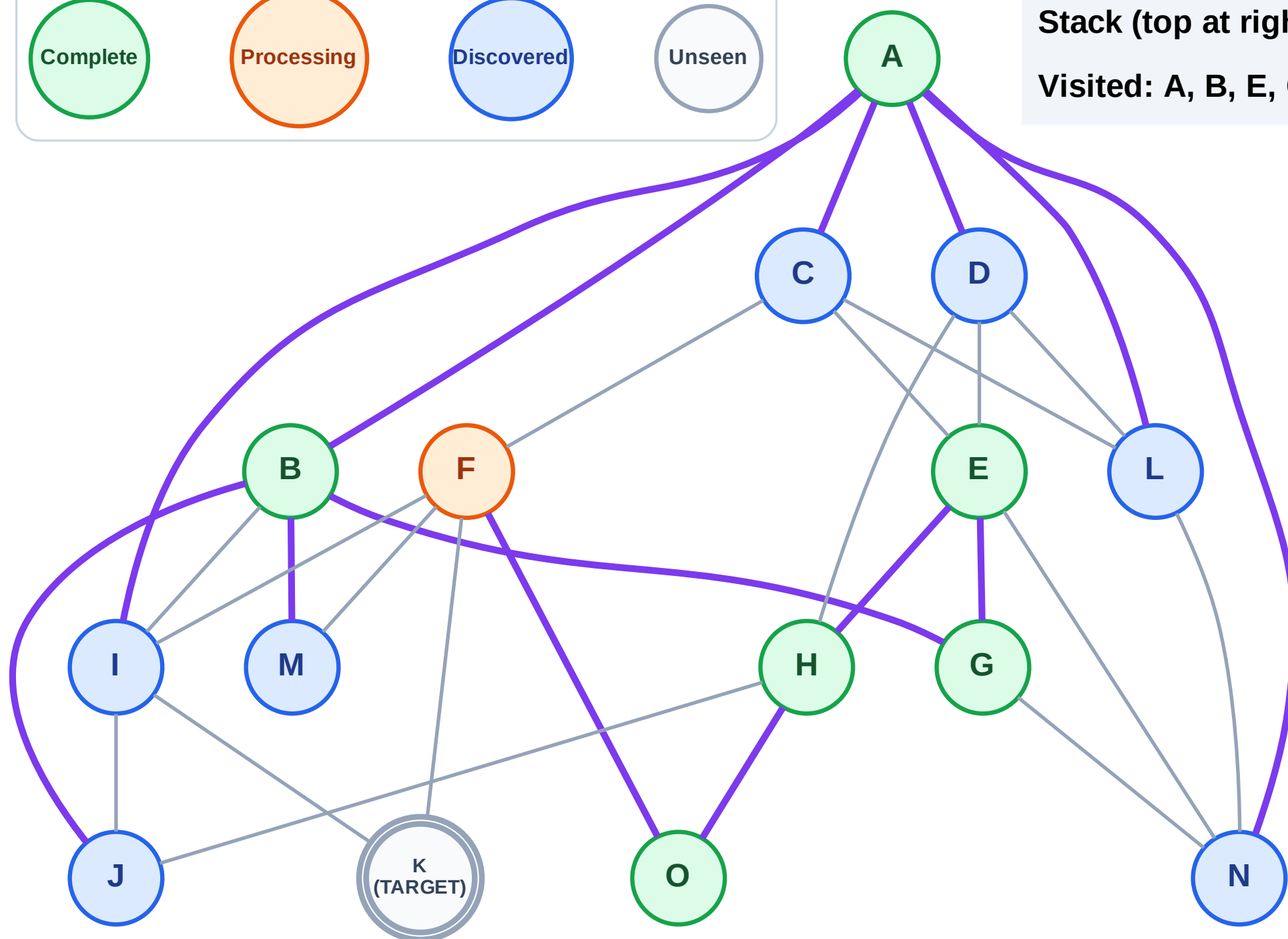
Discovered



Unseen

Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, E, G, H, O



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (7 hops): A → B → G → E → H → O → F → K

Legend



Stack (top at right): N | L | I | D | C | M | J

Visited: A, B, E, F, G, H, K, O

